



Measures of Relative Standing

What is the purpose of relative standing?

- Measures of **relative standing** can be used to compare values from different data sets, or to compare values within the same data set.

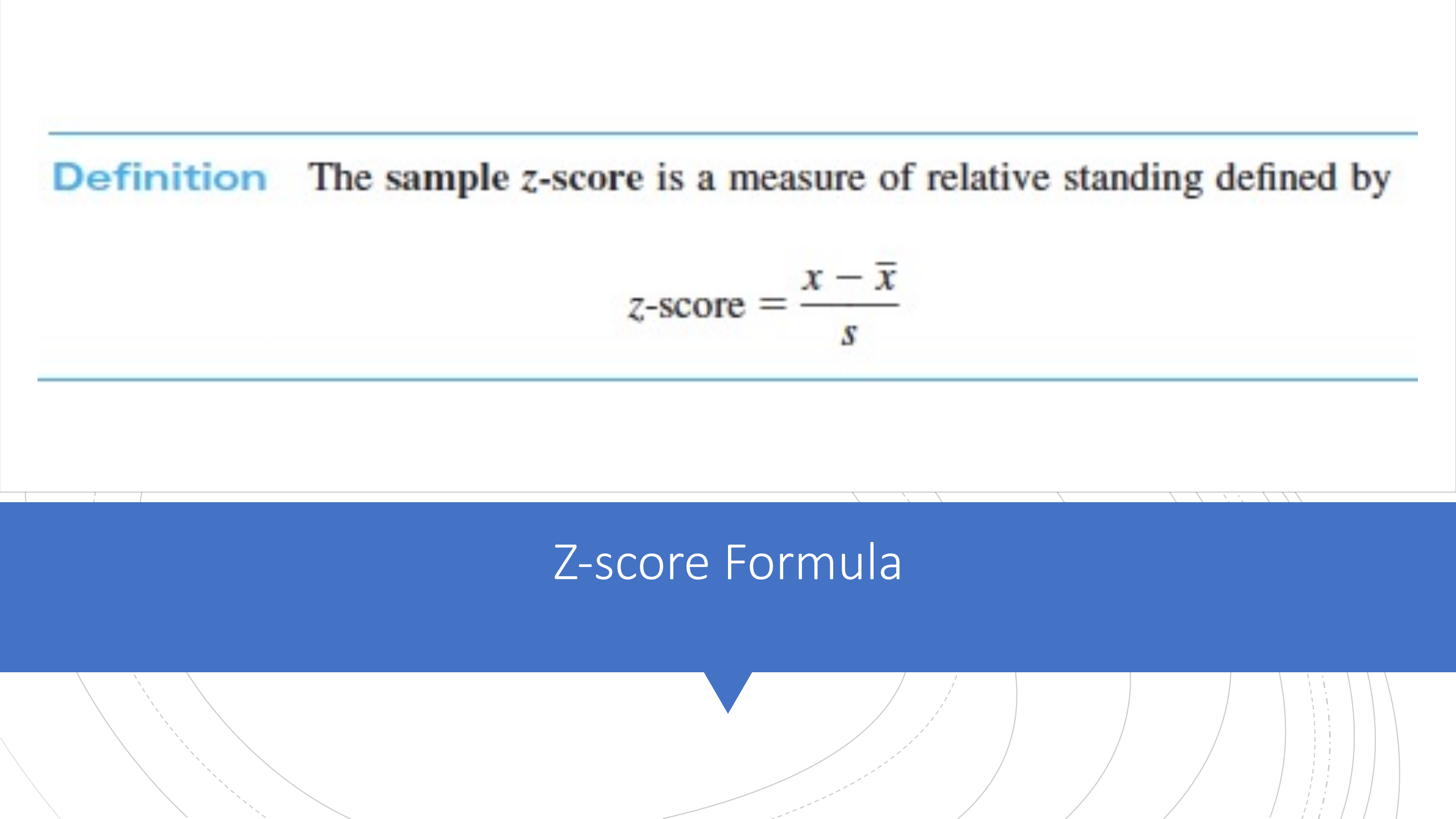
What is the purpose of relative standing?

- Sometimes you need to know the position of one observation relative to others in a set of data.
- For example, if you took an examination with a total of 35 points, you might want to know how your score of 30 compared to the scores of the other students in the class.
- The *mean and standard deviation* of the scores can be used to calculate a *z-score*, which measures the relative standing of a measurement in a data set.

Definition The sample z-score is a measure of relative standing defined by

$$z\text{-score} = \frac{x - \bar{x}}{s}$$

Z-score Formula



Z-score

- A **z-score** measures the distance between an observation and the mean, measured in units of standard deviation.
- For example, suppose that the mean and standard deviation of the test scores (based on a total of 35 points) are 25 and 4, respectively.
- The z-score for your score of 30 is calculated as follows:

$$\text{z-score} = \frac{x - \bar{x}}{s} = \frac{30 - 25}{4} = 1.25$$

Your score of 30 lies 1.25 standard deviations above the mean ($30 = 25 + 1.25 \times 4$).

Purpose of Z-score

- The z-score is a valuable tool for determining whether a particular observation is likely to occur quite frequently or whether it is unlikely and might be considered an *outlier*.

Example

- Consider this sample of $n = 10$ measurements:

1, 1, 0, 15, 2, 3, 4, 0, 1, 3

- The measurement $x = 15$ appears to be unusually large. Calculate the z-score for this observation and state your conclusions.

Solution

- Solution

Solution Calculate $\bar{x} = 3.0$ and $s = 4.42$ for the $n = 10$ measurements. Then the z -score for the suspected outlier, $x = 15$, is calculated as

$$z\text{-score} = \frac{x - \bar{x}}{s} = \frac{15 - 3}{4.42} = 2.71$$

Z scores and Standard Deviations

- Technically, a z-score is the number of standard deviations from the mean value of the reference population (a population whose known values have been recorded like the sample measurements).

For example:

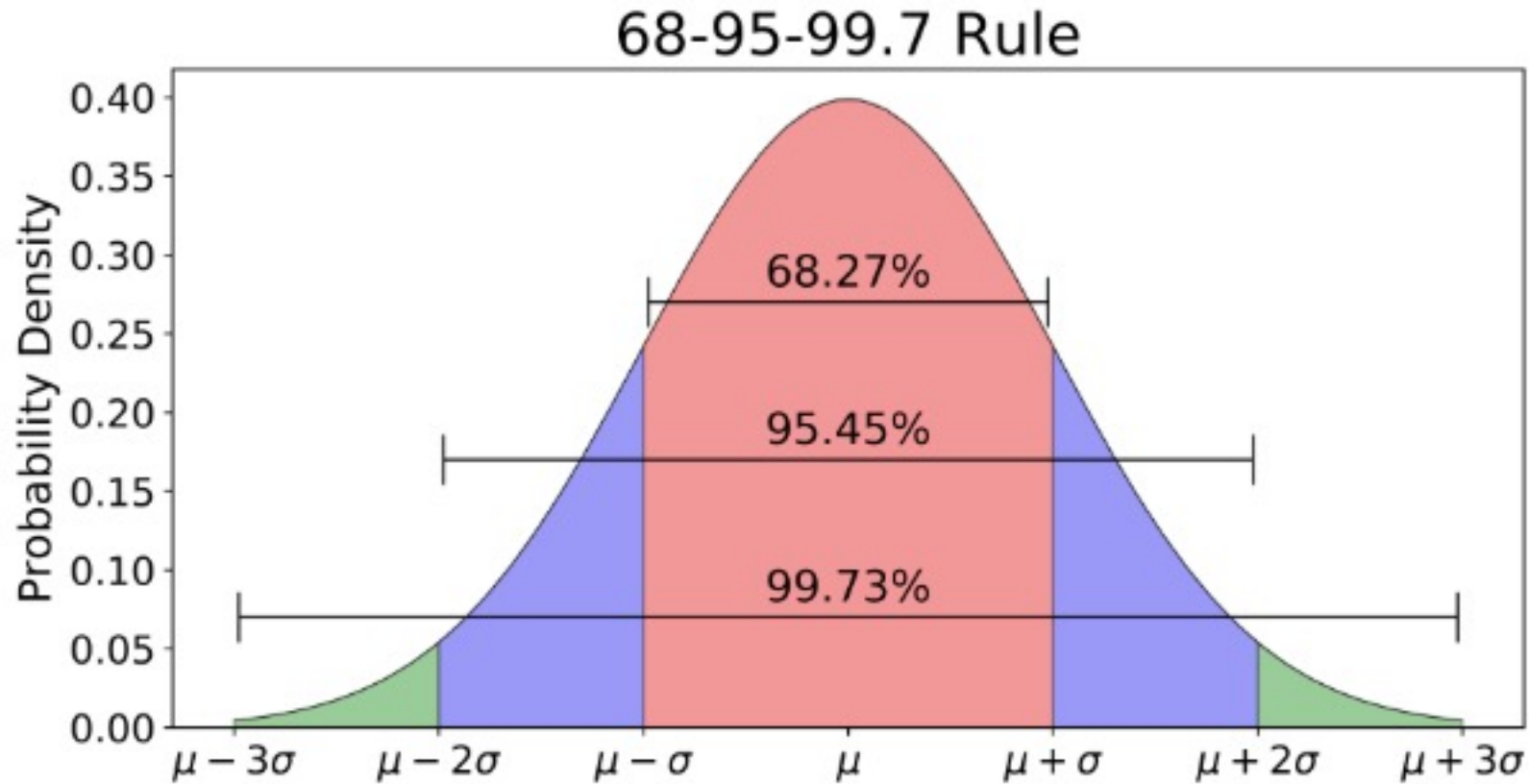
- A z-score of 1 is 1 standard deviation above the mean.
- A score of 2 is 2 standard deviations above the mean.
- A score of -1.8 is -1.8 standard deviations *below* the mean.

Interpreting the Result

The Normal Distribution

- The *normal distribution* is commonly associated with the 68-95-99.7 rule which you can see in the image above. 68% of the data is within 1 standard deviation (σ) of the mean (μ), 95% of the data is within 2 standard deviations (σ) of the mean (μ), and 99.7% of the data is within 3 standard deviations (σ) of the mean (μ).

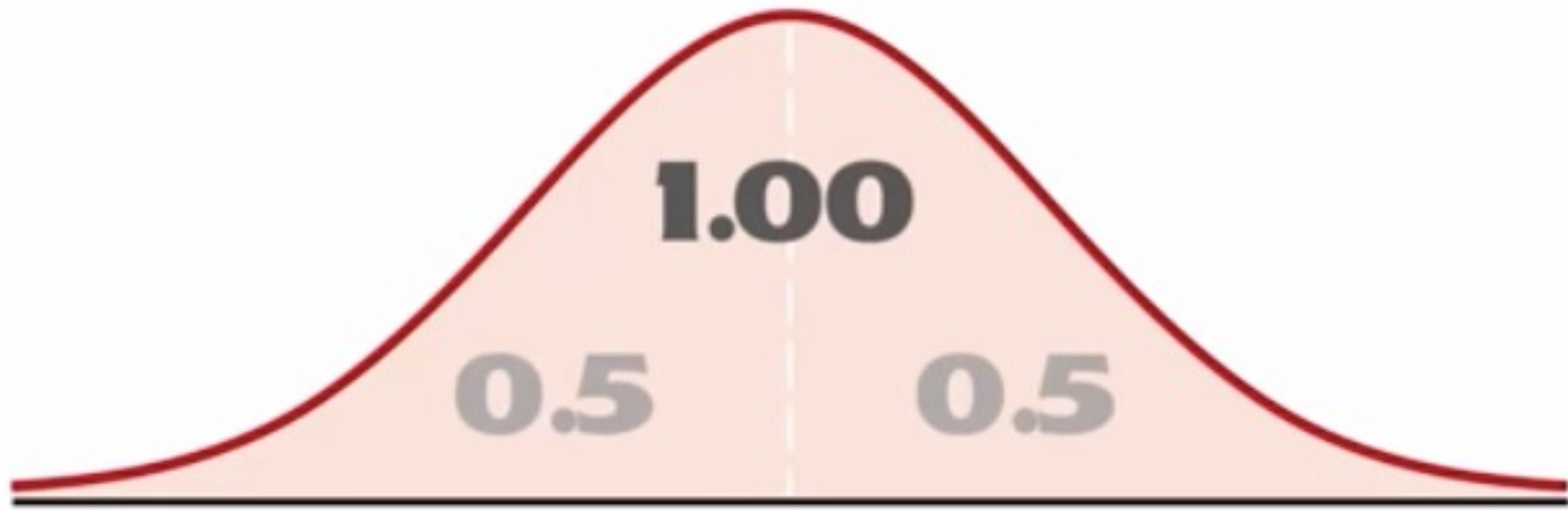
The Normal Distribution



68% of the data is within 1 standard deviation, 95% is within 2 standard deviation, 99.7% is within 3 standard deviations

Normal Distributions

Calculating Probabilities



STANDARD NORMAL DISTRIBUTION

z-distribution

mean $\mu = 0$

std. dev. $\sigma = 1$

STANDARD NORMAL DISTRIBUTION

z-distribution

mean $\mu = 0$

std. dev. $\sigma = 1$

Negative z

Area



LESS THAN
Cumulative

Positive z

Area



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952

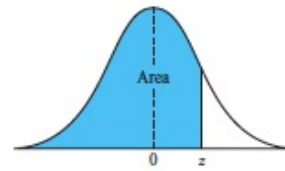


TABLE 3 Areas under the Normal Curve, pages 688–689

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
–3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
–3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
–3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
–3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
–3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
–2.9	.0019	.0018	.0017	.0017	.0016	.0016	.0015	.0015	.0014	.0014
–2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
–2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
–2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
–2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
–2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
–2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
–2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
–2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
–2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
–1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
–1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
–1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
–1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
–1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
–1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0722	.0708	.0694	.0681
–1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
–1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
–1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
–1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
–0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
–0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
–0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
–0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
–0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
–0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
–0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
–0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
–0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
–0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

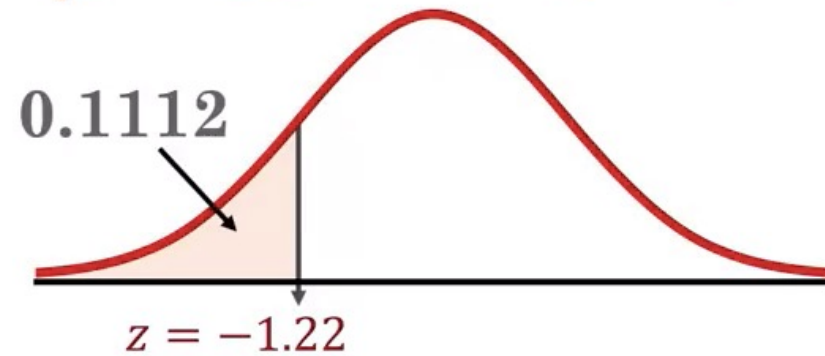
Example

- Scores on exam are normally distributed with a mean of 65 and a standard deviation of 9. Find the percent of the score.
 - a. Less than 54.
 - b. At least 80.
 - c. Between 70 and 86.

Example

$$\mu = 65$$

$$\sigma = 9$$



a) less than 54

b) at least 80

c) between 70 and 86

$$x = 54$$

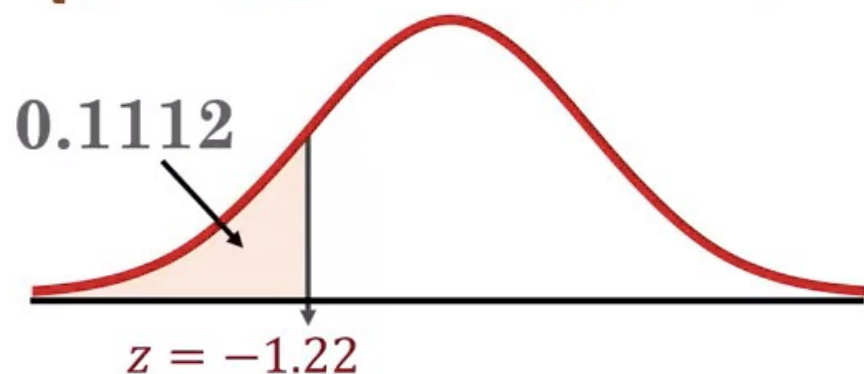
$$z = \frac{x - \mu}{\sigma} = \frac{54 - 65}{9}$$

$$z = -1.22$$

Example

$$\mu = 65$$

$$\sigma = 9$$



a) less than 54 **11.12%**

b) at least 80

c) between 70 and 86

$$x = 54$$

$$z = \frac{x - \mu}{\sigma} = \frac{54 - 65}{9}$$

$$z = -1.22$$

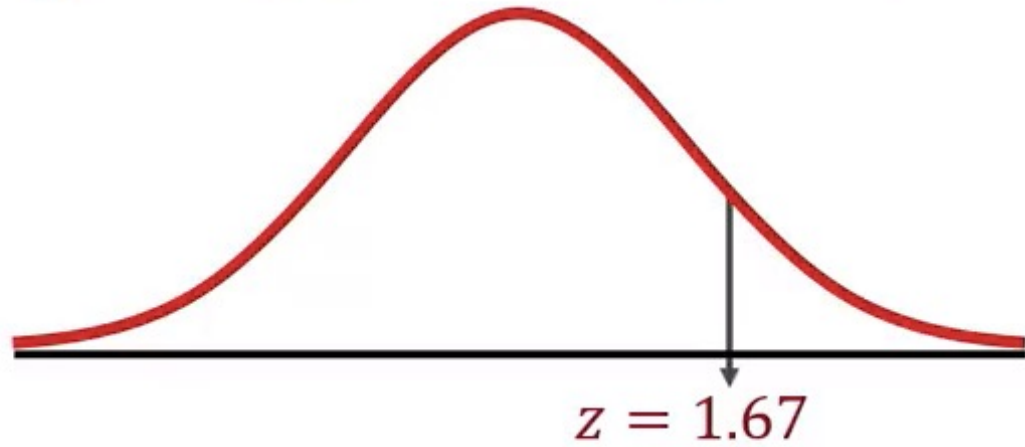
$$\begin{aligned} P(x < 54) \\ &= P(z < -1.22) \\ &= 0.1112 \end{aligned}$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
·	·	·	·	·	·	·	·	·	·	·
·	·	·	·	·	·	·	·	·	·	·
·	·	·	·	·	·	·	·	·	·	·
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
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-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611

Example

$$\mu = 65$$

$$\sigma = 9$$



a) less than 54

b) at least 80

c) between 70 and 86

$$x = 80$$

$$z = \frac{x - \mu}{\sigma} = \frac{80 - 65}{9}$$

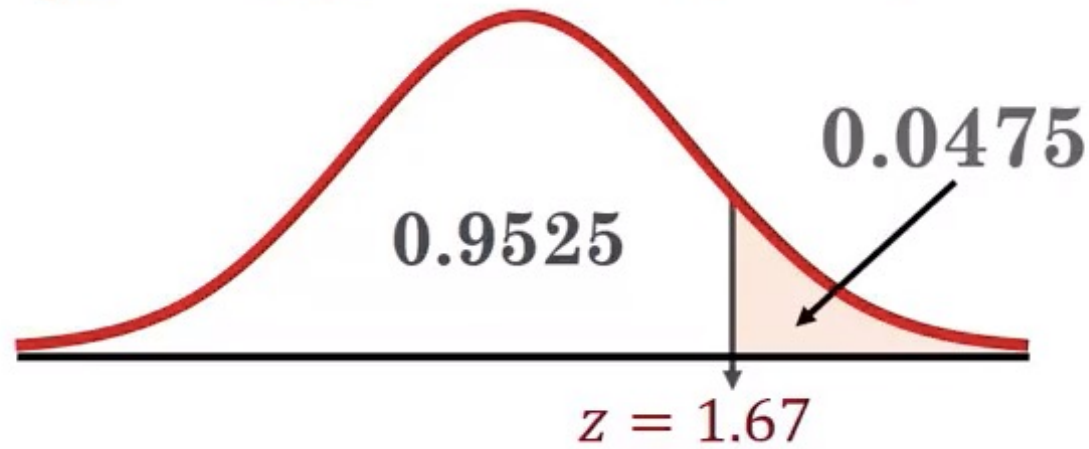
$$z = 1.67$$



Example

$$\mu = 65$$

$$\sigma = 9$$



a) less than 54

b) at least 80 4.75%

c) between 70 and 86

$$x = 80$$

$$z = \frac{x - \mu}{\sigma} = \frac{80 - 65}{9}$$

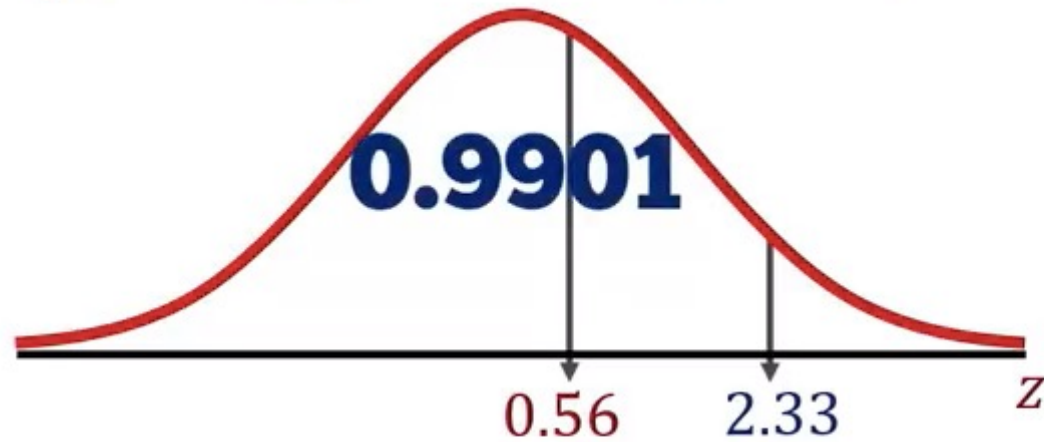
$$z = 1.67$$

$$\begin{aligned} P(x > 80) &= P(z > 1.67) \\ &= 1 - P(z < 1.67) \\ &= 1 - 0.9525 \\ &= 0.0475 \end{aligned}$$

Example

$$\mu = 65$$

$$\sigma = 9$$



a) less than 54

b) at least 80

c) between 70 and 86

$$x = 70$$

$$z = \frac{x - \mu}{\sigma} = \frac{70 - 65}{9} = 0.56$$

$$x = 86$$

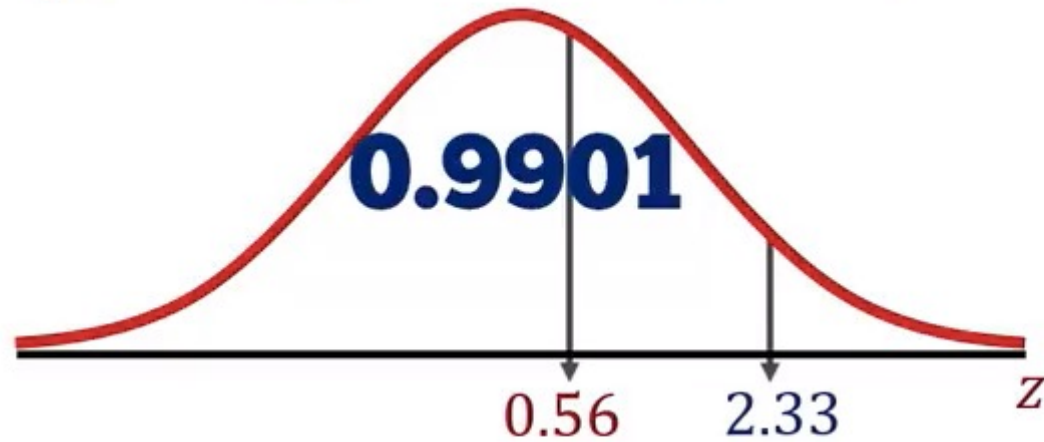
$$z = \frac{x - \mu}{\sigma} = \frac{86 - 65}{9} = 2.33$$

z	0.03
2.3	0.9901

Example

$$\mu = 65$$

$$\sigma = 9$$



a) less than 54

b) at least 80

c) between 70 and 86

$$x = 70$$

$$z = \frac{x - \mu}{\sigma} = \frac{70 - 65}{9} = 0.56$$

$$x = 86$$

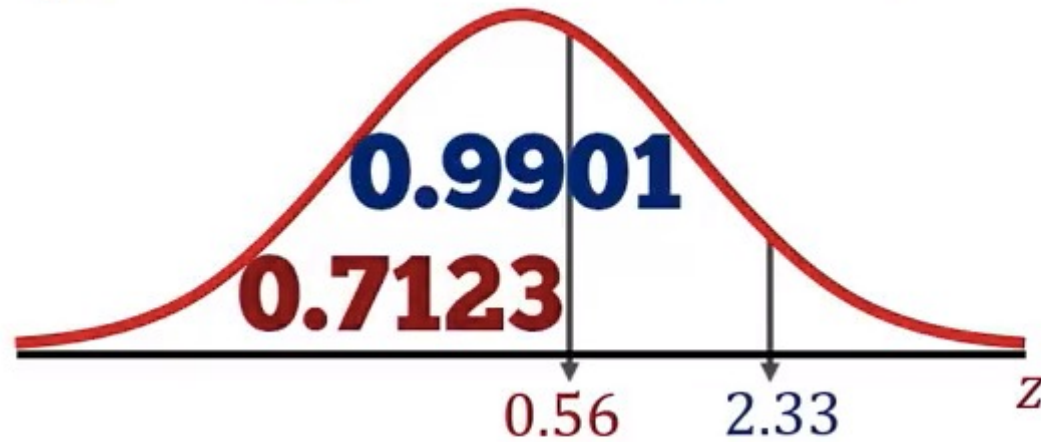
$$z = \frac{x - \mu}{\sigma} = \frac{86 - 65}{9} = 2.33$$

z	0.03	0.06
0.5		0.7123
2.3	0.9901	

Example

$$\mu = 65$$

$$\sigma = 9$$



a) less than 54

b) at least 80

c) between 70 and 86

$$x = 70$$

$$z = \frac{x - \mu}{\sigma} = \frac{70 - 65}{9} = 0.56$$

$$x = 86$$

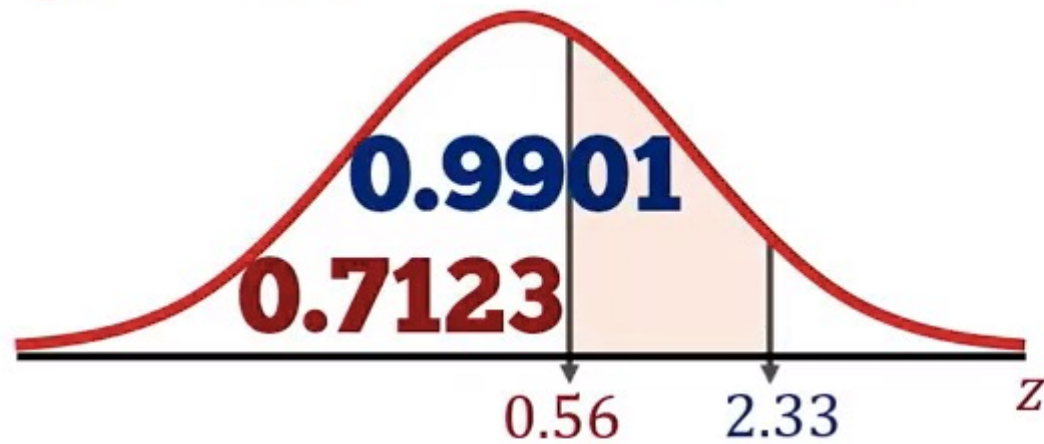
$$z = \frac{x - \mu}{\sigma} = \frac{86 - 65}{9} = 2.33$$

z	0.03	0.06
0.5		0.7123
2.3	0.9901	

Example

$$\mu = 65$$

$$\sigma = 9$$



a) less than 54

b) at least 80

c) between 70 and 86

$$x = 70$$

$$z = \frac{x - \mu}{\sigma} = \frac{70 - 65}{9} = 0.56$$

$$x = 86$$

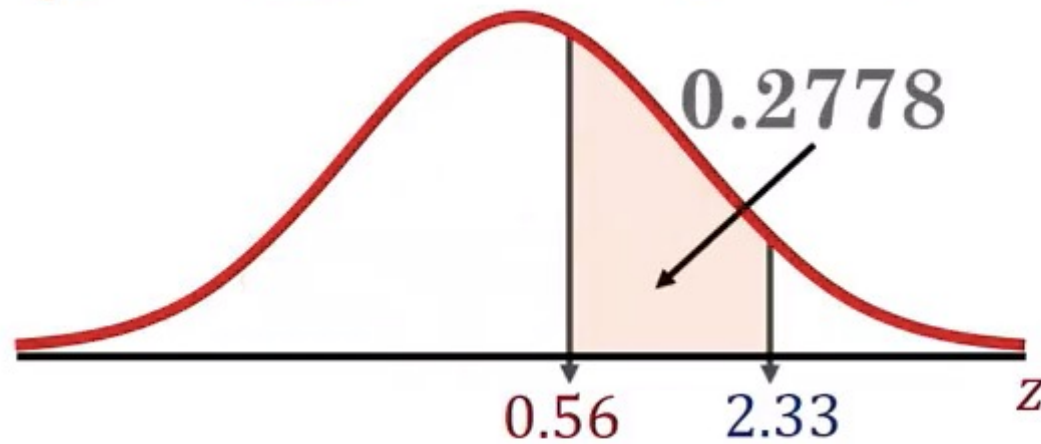
$$z = \frac{x - \mu}{\sigma} = \frac{86 - 65}{9} = 2.33$$

z	0.03	0.06
0.5		0.7123
2.3	0.9901	

Example

$$\mu = 65$$

$$\sigma = 9$$



a) less than 54

b) at least 80

c) between 70 and 86

$$x = 70$$
$$z = \frac{x - \mu}{\sigma} = \frac{70 - 65}{9} = 0.56$$

$$x = 86$$
$$z = \frac{x - \mu}{\sigma} = \frac{86 - 65}{9} = 2.33$$

$$P(70 < x < 86)$$

$$= P(0.56 < z < 2.33)$$

$$= P(z < 2.33) - P(z < 0.56)$$

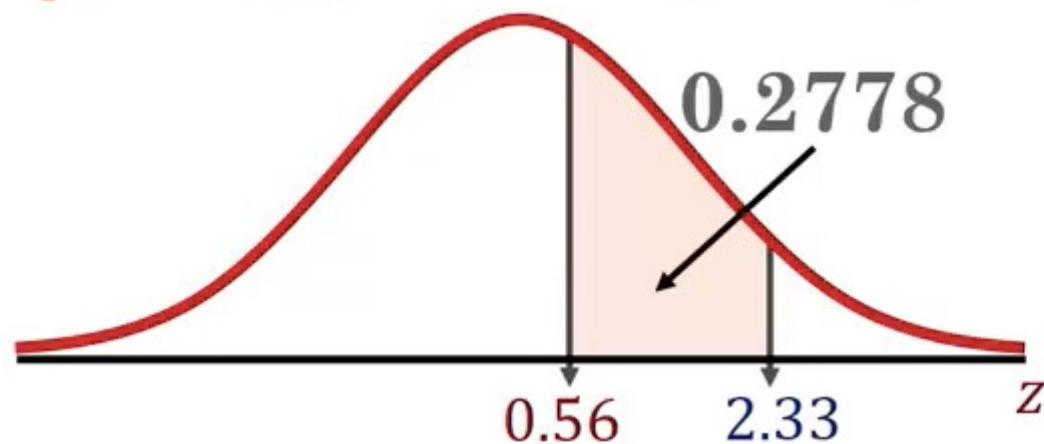
$$= 0.9901 - 0.7123$$

$$= 0.2778$$

Example

$$\mu = 65$$

$$\sigma = 9$$



a) less than 54

b) at least 80

c) between 70 and 86

27.78%

$$x = 70$$

$$z = \frac{x - \mu}{\sigma} = \frac{70 - 65}{9} = 0.56$$

$$x = 86$$

$$z = \frac{x - \mu}{\sigma} = \frac{86 - 65}{9} = 2.33$$

$$P(70 < x < 86)$$

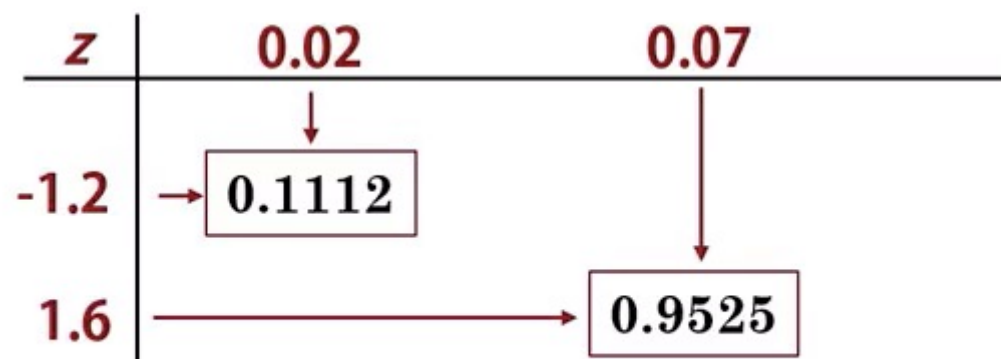
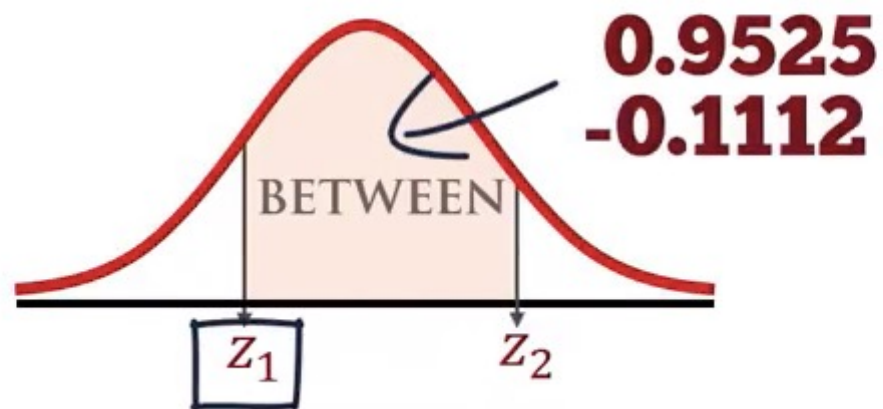
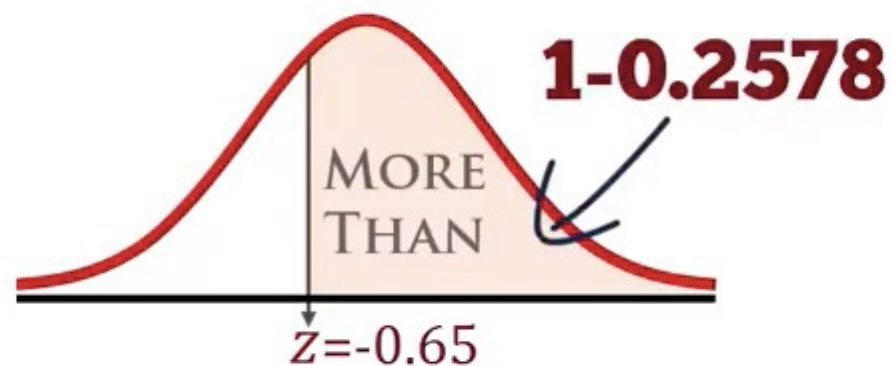
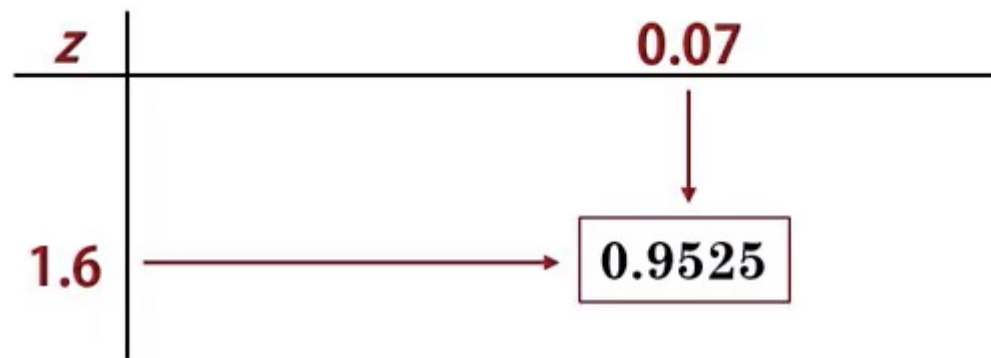
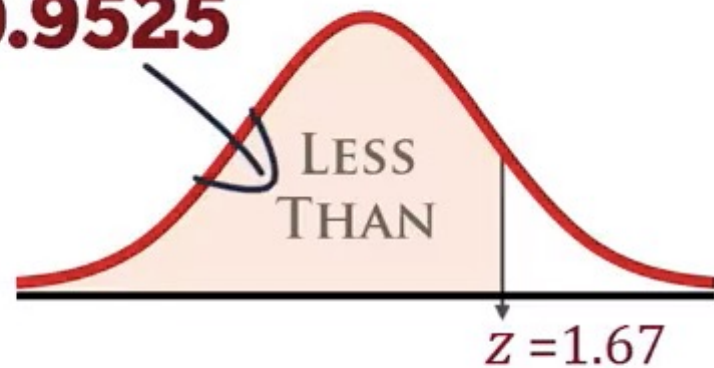
$$= P(0.56 < z < 2.33)$$

$$= P(z < 2.33) - P(z < 0.56)$$

$$= 0.9901 - 0.7123$$

$$= 0.2778$$

0.9525





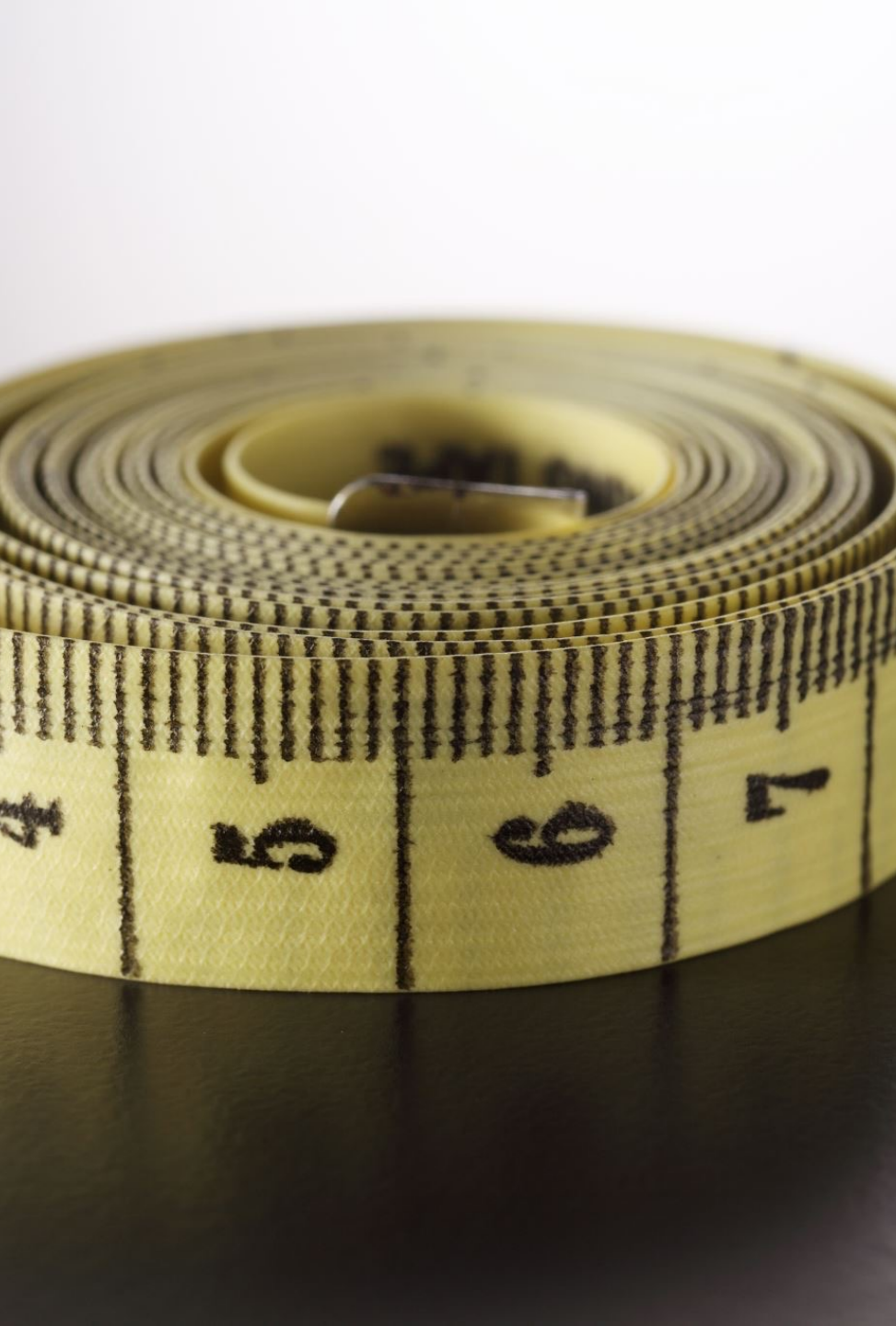
Example 1

- With a height of 67 in., William McKinley was the shortest president of the past century. The presidents of the past century have a mean height of 71.5 in. and a standard deviation of 2.1 in.
 - Convert McKinley's height to a z score.
 - Is the z-score above or below the mean?



Example 2

- Stanford Binet IQ scores have a mean of 100 and a standard deviation of 16. Albert Einstein reportedly had an IQ of 160.
 - Convert Einstein's IQ score to a z score.
 - Is the z-score above or below the mean?
 - Find the percent of the IQ score based on the z-score table.
 - If we consider “usual” IQ scores to be those that convert to z scores between -2 and 2, is Einstein's IQ usual or unusual?



Example 3

- Men have heights with a mean of 176 cm and a standard deviation of 7 cm. Charles Darwin had a height of 182 cm.
 - Convert Darwin's height to a z-score.
 - Is the z-score above or below the mean?
 - Find the percent of Darwin's height based on the z-score table.



Solve

- Human body temperatures have a mean of 98.20°F and a standard deviation of 0.62°F . Convert the given temperatures to z scores (2pts each) and find the percent of the temperature based on the z-score table.
- 97.5°F
- 98.60°F

Percentile

- A percentile is a value which a certain percentage of observations lie.

Example of a data set:

2, 2, 3, 4, 5, 5, 5, 6, 7, 8, 8, 8, 8, 8, 9, 9, 10, 11, 11, 12

What is the ranking of 10?